AMRSEC.STUDY

ELECTRONIC DEVICES AND CIRCUITS

DETAILED TOPIC-WISE 10-MARK ANSWERS

JNTUH B.Tech CSE • R25 Regulations • EC104ES

Designed for: concept clarity, descriptive answers, circuit operation, derivations, worked numericals and last-minute revision.

Independent educational study material • Not an official university publication

How to use this guide

This edition expands major syllabus topics into structured, exam-oriented answers. For a 10-mark response, adapt the length to the question and the time available: define the concept, explain the principle, draw a labelled diagram, describe operation step by step, add equations or characteristics where relevant, and finish with applications or a conclusion.

Diagrams are described as drawing instructions where a reliable vector schematic is not embedded. In your exam, draw them neatly with correct terminal labels and arrows. Formula models use stated assumptions; actual device behavior depends on operating conditions and datasheets.

Syllabus alignment

The unit organization follows the JNTUH R25 CSE EC104ES Electronic Devices and Circuits syllabus: Unit I diode characteristics and applications; Unit II BJT; Unit III BJT biasing; Unit IV transistor amplifiers; Unit V special diodes, FETs and advanced devices. Use your faculty's prescribed text and class notes if a local notation convention differs.

10-mark answer framework

• Definition and purpose (about 1 mark).

• Principle / construction and labelled diagram (2 marks).

• Step-by-step working or theory (3 marks).

• Equations, derivation, characteristics or numerical analysis (2–3 marks).

• Applications, advantages/limitations and conclusion (1 mark).

Contents

• Unit I — Diode characteristics and applications

• Unit II — Bipolar junction transistor

• Unit III — BJT biasing and stabilization

• Unit IV — Transistor amplifiers

• Unit V — Special diodes, FETs and advanced devices

• Formula bank, numerical practice and final revision

UNIT I — DIODE CHARACTERISTICS AND APPLICATIONS

1. Intrinsic and extrinsic semiconductors

Introduction — Semiconductors have electrical conductivity between conductors and insulators. Their conductivity can be controlled by temperature, light, electric fields and impurity doping.

Intrinsic semiconductor

An intrinsic semiconductor is a pure material such as silicon or germanium. At absolute zero, its valence band is full and conduction band is nearly empty. At ordinary temperatures, thermal energy can break some covalent bonds, producing electron-hole pairs. The number of free electrons equals the number of holes, so n = p = n_i. Both carrier types contribute to current.

Extrinsic semiconductor

Doping introduces a controlled impurity concentration. Pentavalent donor atoms such as phosphorus provide an extra electron and form N-type material; trivalent acceptors such as boron create holes and form P-type material. N-type material has electrons as majority carriers and holes as minority carriers. P-type material has holes as majority carriers and electrons as minority carriers.

Conductivity and importance

Conductivity can be expressed as $\sigma = q(n\mu_n + p\mu_p)$, where q is electron charge magnitude, n and p are carrier concentrations, and μ_n and μ_p are mobilities. Doping changes carrier concentration strongly, allowing diodes, BJTs and MOSFETs to be engineered.

How to present in an exam

Draw a simple band diagram for intrinsic material and a donor/acceptor energy-level sketch if asked. Define majority/minority carriers, distinguish donor and acceptor doping, and explain why doping changes conductivity.

> **Exam conclusion / points to remember:** Intrinsic: n=p=n_i. N-type: electrons are majority carriers. P-type: holes are majority carriers. Doping changes conductivity without making the material a metal.

Practice: Explain intrinsic and extrinsic semiconductors, doping, majority/minority carriers and conductivity.

2. PN junction formation and depletion region

Introduction — A PN junction is formed when P-type and N-type regions exist in the same semiconductor crystal. Its behavior is the foundation of a diode.

Formation

Immediately after the regions meet, electrons diffuse from the N side to the P side because electron concentration is higher on the N side. Holes diffuse in the opposite direction. Near the junction, electrons and holes recombine.

Depletion region and electric field

Recombination leaves fixed positive donor ions on the N side and fixed negative acceptor ions on the P side. The region contains very few mobile carriers and is called the depletion region. These fixed ions create an electric field and a built-in potential barrier that opposes further diffusion.

Equilibrium

At thermal equilibrium, diffusion current and drift current balance. There is no net current through the junction without external bias. The barrier depends on semiconductor material, doping and temperature; it is not a battery-like source that continuously supplies energy.

Forward and reverse bias

Forward bias reduces the barrier and depletion width, allowing injected carriers to cross the junction. Reverse bias increases the barrier and depletion width, leaving only a small leakage current until breakdown.

Exam conclusion / points to remember: Explain diffusion, recombination, fixed ions, electric field and equilibrium in sequence. A labelled P-region / depletion region / N-region sketch earns clarity.

3. Diode I–V characteristics and diode models

Introduction — The current-voltage curve describes how diode current responds to the voltage across the junction. It is used to predict whether a diode conducts in a circuit.

Forward region

At low forward voltage, current is small. As the junction voltage increases, injected carrier concentration rises exponentially and current increases rapidly. A silicon diode is often approximated with a 0.7 V drop for elementary circuit calculations, but the real drop depends on current, temperature and device construction.

Reverse region

Under reverse bias, the diode carries a small reverse saturation current due to minority carriers. If reverse voltage reaches breakdown, current increases sharply. Zener and avalanche mechanisms can cause breakdown; a suitable current-limiting circuit is essential.

Shockley equation

EQ:I_D = I_S [exp(V_D/(nV_T)) – 1]

Parameter definitions

I_D is diode current, I_S is reverse saturation current, V_D is junction voltage, n is ideality factor and V_T=kT/q is thermal voltage. Near room temperature V_T is approximately 25.9 mV. The equation is an idealized model and does not capture all high-current series-resistance effects.

Practical models

Ideal model: on-state short circuit, off-state open circuit. Constant-drop model: forward diode drop approximated by Vγ. Piecewise-linear model: V_D≈Vγ+I_Dr_f. Choose the simplest model that meets the problem's required accuracy.

Exam conclusion / points to remember: Label both axes and mark forward conduction, reverse leakage and breakdown. State which diode model you use before solving a circuit.

Practice: Draw and explain the diode I–V characteristics. Compare ideal, constant-drop and piecewise-linear models.

4. Diode resistance and junction capacitance

Introduction — Diode resistance and capacitance describe its response around a DC operating point and help explain high-frequency behavior.

DC or static resistance

At a selected operating point, the ratio of diode voltage to diode current gives the static resistance:

Static resistance

EQ:R_D = V_D / I_D

Dynamic or small-signal resistance

The dynamic resistance is the local slope of the I–V curve. Under the ideal exponential model in forward bias, r_d≈nV_T/I_D. It decreases as forward current increases. This is a local, small-signal quantity and should not be confused with V_D/I_D.

Junction capacitance

The depletion region separates charge and behaves like a capacitor. Reverse bias widens the depletion region, usually reducing junction capacitance. A simplified relation is C_j≈εA/W, where ε is permittivity, A is junction area and W is depletion width.

Diffusion capacitance

In forward bias, injected carriers are stored in the neutral regions. The change in stored charge with voltage gives diffusion capacitance. It can dominate junction capacitance under strong forward conduction and affects switching speed.

Exam conclusion / points to remember: Static resistance uses V/I; dynamic resistance uses a slope. Junction capacitance is associated with depletion charge; diffusion capacitance is associated with injected charge.

5. Half-wave rectifier

Introduction — A rectifier converts AC into a unidirectional voltage. The half-wave rectifier is the simplest type and uses one diode with a load resistor.

Circuit and positive half-cycle

Connect the AC source, diode and load in series. During the half-cycle that forward-biases the diode, current flows through the load. For an ideal diode, the load voltage follows the positive half of the input sinusoid.

Negative half-cycle

During the opposite half-cycle, the diode is reverse biased and ideally blocks current. The load voltage is approximately zero. The output therefore contains separated pulses rather than a smooth DC level.

Average output derivation

For v_o=V_m sinθ from 0 to π and zero from π to 2π, the average over one full cycle is (1/2π)∫■^π V_m sinθ dθ = V_m/π.

Key ideal results

EQ:V_DC = V_m/π; I_DC = I_m/π

Performance and limitations

Ripple frequency equals supply frequency f. Ideal maximum rectification efficiency is approximately 40.6%; practical efficiency is lower. The output has large ripple and transformer utilization is poor, so half-wave rectification is used mainly in simple, low-power applications.

Exam conclusion / points to remember: Draw the series circuit and input/output waveforms. Show that only one half-cycle reaches the load. Mention diode drop in practical calculations.

Practice: Explain the operation, waveform and performance of a half-wave rectifier.

6. Full-wave rectifiers: centre-tap and bridge

Introduction — A full-wave rectifier uses both input half-cycles, producing a higher average output and ripple at twice the supply frequency.

Centre-tapped design

A centre-tapped secondary and two diodes are arranged so that one diode conducts on each half-cycle. The load current flows in the same direction during both half-cycles. Each diode must withstand a relatively high peak inverse voltage.

Bridge design

Four diodes form a bridge. Two diodes conduct during each half-cycle, steering current through the load in the same direction. A centre-tapped transformer is not required, but two diode drops occur in the conducting path.

Average output

For an ideal full-wave rectified sine wave, the average is the average of $|V_m \sin\theta|$ over a full cycle:

Equation

EQ:$V_{DC} = 2V_m/\pi$; $I_{DC} = 2I_m/\pi$

Comparison

Full-wave ripple frequency is 2f. Ideal maximum rectification efficiency is about 81.2%. The bridge uses four diodes but has no centre tap; the centre-tapped version uses two diodes but needs a centre-tapped secondary. Practical output is reduced by diode drops and source resistance.

> **Exam conclusion / points to remember:** Always state whether V_m refers to the peak voltage across a half-secondary or across the complete secondary. This matters when comparing transformer configurations.

Practice: Compare centre-tapped and bridge rectifiers with circuits, waveforms and performance.

7. Capacitor filter and ripple

Introduction — The output of a rectifier is pulsating DC. A filter reduces the AC variation, called ripple, to make the output more nearly constant.

Operation

A capacitor is connected in parallel with the load. Near each rectified peak, the diode conducts and charges the capacitor close to the peak voltage. When the source voltage falls below the capacitor voltage, the diode stops conducting and the capacitor discharges through the load.

Ripple behavior

The output falls gradually between charging peaks. Increasing capacitance reduces the rate of voltage fall. Increasing load current increases discharge and ripple. A full-wave rectifier refreshes the capacitor twice per AC cycle, so ripple is lower than for half-wave rectification under otherwise similar conditions.

Approximate derivation

For small ripple, capacitor charge change $\Delta Q \approx I_L \Delta t$. Since $\Delta Q = C\Delta V$, ripple $\Delta V \approx I_L \Delta t/C$. With ripple frequency f_r, $\Delta t \approx 1/f_r$, giving:

Equation

EQ:$V_{r(pp)} \approx I_L/(f_r C)$

Design limitations

This approximation assumes small ripple and a reasonably constant load current. Real circuits also depend on transformer resistance, diode conduction angle, capacitor ESR and load variation. Choose capacitor voltage rating and ripple-current rating appropriately.

> **Exam conclusion / points to remember:** Define ripple as the residual AC variation on the DC output. State $f_r = f$ for half-wave and $f_r = 2f$ for full-wave rectification.

8. Clippers and clampers

Introduction — Clippers and clampers shape a waveform using diodes, but they perform different jobs.

Clipper principle

A clipper removes or limits a selected part of the input waveform. A series clipper places the diode in series with the load; a shunt clipper places the diode across the output path. Diode orientation determines which polarity is clipped. A bias source can set a clipping level other than zero.

Clipper operation

When the diode is forward biased, it conducts and the output is either passed or shunted depending on the topology. When reverse biased, it blocks and the output follows a different path. Analyse each input half-cycle separately and include the diode's practical forward drop if required.

Clamper principle

A clamper shifts the whole waveform upward or downward by adding a DC component. It commonly uses a diode, capacitor and resistor. During one half-cycle the capacitor charges; during the other, it retains most of its charge and adds its voltage to the input.

Time constant

For good clamping, the RC time constant should usually be large compared with the signal period so the capacitor discharges only slightly during a cycle. The exact output depends on diode direction, initial charge, load and any bias source.

Exam presentation

Draw the circuit and input/output waveforms. State which half-cycle charges the capacitor or conducts the diode, then explain how the resulting output is clipped or shifted.

Exam conclusion / points to remember: Clipper = waveform amplitude limiting. Clamper = DC-level shifting. Do not describe a clamper as simply cutting off one half-cycle.

9. Zener diode voltage regulator

Introduction — A Zener diode is designed to operate in reverse breakdown and can maintain an approximately constant voltage across a load.

Circuit arrangement

Connect a series resistor from the supply to the output node. Connect the Zener diode reverse biased across the load, with the cathode at the positive output node for a positive regulated voltage. The resistor limits current.

Operation

If the supply rises, additional current flows through the resistor and mostly through the Zener, helping keep the output near V_Z. If load current rises, Zener current falls to supply more of the available current to the load. Regulation works only while Zener current stays between its minimum knee current and maximum safe current.

Design equation

EQ:$R_S = (V_S - V_Z)/(I_L + I_Z)$

Power check

EQ:$P_Z = V_Z I_Z$

Worked example

Let $V_S=12$ V, $V_Z=5.1$ V, $R_S=330\ \Omega$ and load current $I_L=10$ mA. Series current is $I_R=(12-5.1)/330\approx20.9$ mA. Zener current is $I_Z\approx20.9-10=10.9$ mA. Zener power is $P_Z\approx5.1\times0.0109=55.6$ mW.

Design checks

Check minimum-supply/maximum-load conditions to ensure the Zener still conducts enough current. Check maximum-supply/minimum-load conditions to ensure current and power remain below ratings. Include tolerance and temperature margins in a real design.

Exam conclusion / points to remember: A Zener is not a perfect voltage source. Regulation requires a suitable series resistor, a valid current range and adequate power rating.

Practice: Explain Zener regulation and calculate current and power for a given circuit.

UNIT II — BIPOLAR JUNCTION TRANSISTOR

1. BJT construction and transistor action

Introduction — A BJT is a three-terminal semiconductor device in which both electrons and holes participate in conduction. It can amplify signals or act as a switch.

Construction

The emitter is heavily doped to inject carriers. The base is extremely thin and lightly doped so most injected carriers cross it without recombining. The collector is moderately doped and physically designed to collect carriers and dissipate heat. NPN and PNP transistors differ in the order of semiconductor regions and carrier directions.

Active operation

For an NPN transistor in forward-active operation, the base-emitter junction is forward biased and the base-collector junction is reverse biased. Electrons move from emitter into base; most diffuse across the thin base and are swept into the collector. A small fraction recombines, creating base current.

Current relationship

EQ:$I_E = I_C + I_B$

Amplification

Because base recombination current is relatively small, a small base current controls a much larger collector current. In active operation, $I_C \approx \beta I_B$. This relation is approximate and depends on operating point and device properties.

Operating regions

Cutoff: both junctions are effectively reverse biased and collector current is small. Active: base-emitter forward, base-collector reverse; useful for amplification. Saturation: both junctions forward biased; used for switching ON. Reverse-active operation is possible but rarely used because performance is poor.

> **Exam conclusion / points to remember:** Label emitter, base, collector and doping levels. Explain why the thin base is crucial to transistor action.

2. BJT current gains and numerical relations

Introduction — Current gain expresses the relationship between transistor terminal currents and is central to analysing BJT circuits.

Common-base gain α

α is the ratio of collector current to emitter current:

Equation

EQ:$\alpha = I_C/I_E$

Common-emitter gain β

β is the ratio of collector current to base current:

Equation

EQ:$\beta = I_C/I_B$

Derivation

Since $I_E=I_C+I_B$ and $I_C=\alpha I_E$, then $I_B=I_E–I_C=I_E(1–\alpha)$. Therefore $\beta=I_C/I_B=\alpha/(1–\alpha)$. Rearranging gives $\alpha=\beta/(\beta+1)$.

Worked example

If $\alpha=0.98$, $\beta=0.98/(1–0.98)=49$. If $I_B=40\ \mu A$, $I_C \approx \beta I_B=1.96$ mA and $I_E \approx I_C+I_B=2.00$ mA.

Practical note

DC current gain varies from device to device and with collector current, temperature and collector-emitter voltage. Use a datasheet range or a stated β in numerical questions; do not assume one universal gain.

Exam conclusion / points to remember: EQ:$\beta = \alpha/(1-\alpha)$; $\alpha = \beta/(\beta+1)$

Practice: Derive the α–β relationship and solve a transistor current calculation.

3. Common-base configuration

Introduction — In the common-base configuration, the base terminal is common to input and output circuits. The input is applied between emitter and base and the output is taken between collector and base.

Operation

In active operation the emitter-base junction is forward biased and collector-base junction reverse biased. Emitter current injects carriers; most are collected, so collector current is close to emitter current.

Characteristics

Input characteristics plot emitter current I_E against emitter-base voltage V_{EB} for fixed collector-base voltage. Output characteristics plot collector current I_C against collector-base voltage V_{CB} for fixed emitter current. Input resistance is low; output resistance is high.

Gain and phase

Current gain $\alpha = I_C/I_E$ is less than but usually near unity. Voltage gain can be high because the output resistance is high. CB does not show the usual CE 180° voltage phase inversion.

Applications and limitations

CB can be useful at high frequencies and for matching a low-resistance source to a high-resistance load. Its low input resistance makes it less suitable as a general voltage-amplifier input stage.

Exam conclusion / points to remember: Remember: low input resistance, high output resistance, current gain below unity and no typical CE phase inversion.

4. Common-emitter configuration

Introduction — The common-emitter configuration is widely used because it can provide both current gain and voltage gain.

Connections and operation

The emitter is common to input and output. The input signal is applied between base and emitter; output is measured between collector and emitter. In active operation, a change in base current produces a larger change in collector current.

Input characteristic

Plot I_B versus V_{BE} at fixed V_{CE}. It resembles a forward-biased PN junction because the base-emitter junction is forward biased.

Output characteristic

Plot I_C versus V_{CE} for several fixed I_B values. At low V_{CE} the transistor is in saturation; at moderate V_{CE} it operates in active region; at $I_B \approx 0$ it is near cutoff. In active region I_C is approximately βI_B.

Phase inversion

If base current increases, collector current increases and the drop across R_C increases. Collector voltage falls. Thus the collector output voltage is approximately 180° out of phase with the input voltage.

Applications

CE stages are common in voltage amplification and switching. Biasing is essential to keep the transistor in the desired region and avoid clipping.

Exam conclusion / points to remember: Draw both input and output characteristics, label the fixed parameter for each curve, and mark cutoff, active and saturation regions.

5. Common-collector configuration (emitter follower)

Introduction — The common-collector circuit takes output from the emitter while the collector is common to the input and output AC circuits.

Operation

A change in base voltage causes the emitter voltage to follow it, with a base-emitter junction drop between them. The output signal is approximately in phase with the input.

Characteristics

It has high input resistance, low output resistance and voltage gain slightly less than unity. Current gain is high because emitter current includes both collector and base currents.

Why it is useful

The emitter follower can accept a signal from a high-resistance source and drive a lower-resistance load without large voltage loss. It is therefore useful as a buffer and impedance-matching stage.

Limitations

It does not provide substantial voltage gain. The output swing is limited by transistor bias and supply rails, and the base-emitter voltage varies with current and temperature.

Exam conclusion / points to remember: CC = emitter follower: high input resistance, low output resistance, near-unity voltage gain, no typical phase inversion.

6. Comparison of CB, CE and CC

Introduction — The three basic BJT configurations use different common terminals and therefore trade input resistance, output resistance and gain.

Comparison

CB has low input resistance and current gain near unity; CE has moderate input resistance with both current and voltage gain and phase inversion; CC has high input resistance, low output resistance and near-unity voltage gain.

How to choose

Use CE when voltage amplification is required. Use CC when buffering or impedance transformation is needed. Use CB when a low input resistance or high-frequency performance is useful. Actual values depend on bias, load and device.

Exam method

Present a table with common terminal, input/output resistance, current gain, voltage gain, phase relation and application. Avoid claiming fixed numerical values unless given by the circuit.

Exam conclusion / points to remember: An accurate comparison table is often the clearest way to answer this topic.

7. BJT h-parameters from characteristics

Introduction — The hybrid or h-parameter model describes a transistor as a two-port network using small changes around a DC operating point.

CE two-port equations

EQ:v_i = h_ie i_i + h_re v_o

Second equation

EQ:i_o = h_fe i_i + h_oe v_o

Meaning of parameters

h_ie is input resistance in ohms; h_re is reverse voltage ratio; h_fe is forward current gain; h_oe is output admittance in siemens. Parameters depend on configuration, operating point and frequency.

Determining values

At fixed output voltage, h_ie≈Δv_i/Δi_i and h_fe≈Δi_o/Δi_i. At fixed input current, h_re≈Δv_i/Δv_o and h_oe≈Δi_o/Δv_o. For CE transistor characteristics these are found from local increments on the input and output curves.

Procedure

Choose the operating point; identify the relevant curve; select two nearby points while keeping the required variable fixed; calculate the ratio of changes; include units for resistance/admittance parameters. Large intervals can give inaccurate small-signal estimates.

Exam conclusion / points to remember: Use increments around the operating point, not arbitrary distant points. State the configuration because h-parameters change with configuration.

UNIT III — BJT BIASING AND STABILIZATION

1. Need for biasing and operating point

Introduction — Biasing applies DC voltages and currents to establish a transistor's quiescent operating point before an AC signal is applied.

Q-point

The quiescent point is the DC collector current I_{CQ} and collector-emitter voltage V_{CEQ} with no input signal. For a linear amplifier, the Q-point should allow the required signal swing without entering cutoff or saturation.

Why it matters

Transistor gain varies with temperature and device-to-device β variation. A poorly chosen bias point can cause clipping, distortion or excessive power dissipation. Biasing aims to make operation predictable and reasonably stable.

Load line

For a CE circuit with supply V_{CC} and collector resistor R_C, KVL gives $V_{CC}=I_C R_C+V_{CE}$, so $I_C=(V_{CC}-V_{CE})/R_C$. The line connects $(V_{CE}=V_{CC},I_C=0)$ and $(V_{CE}=0,I_C=V_{CC}/R_C)$. The Q-point is where the load line meets the curve corresponding to the base bias.

Power consideration

Approximate transistor dissipation is $P_D \approx V_{CE} I_C$. The Q-point and signal swing must respect maximum power, voltage and current ratings.

> **Exam conclusion / points to remember:** Biasing establishes the DC operating point; the load line expresses the external circuit constraint.

2. Fixed-bias circuit and Q-point calculation

Introduction — Fixed bias is a simple transistor-bias circuit in which a base resistor is connected to the supply.

Base current

For an NPN transistor with emitter grounded, $I_B \approx (V_{CC}-V_{BE})/R_B$. This assumes the base-emitter junction drop is known and the circuit is in the expected operating region.

Collector current and voltage

In active region, $I_C \approx \beta I_B$. Collector voltage is $V_C=V_{CC}-I_C R_C$ and $V_{CE} \approx V_C$ if the emitter is grounded. If the calculated collector current would require V_{CE} below a plausible saturation voltage, the active-region calculation is invalid and the transistor is saturated.

Worked example

Given $V_{CC}=12$ V, $R_B=240\ k\Omega$, $R_C=2\ k\Omega$, $V_{BE}=0.7$ V and $\beta=100$: $I_B=(12-0.7)/240k\Omega=47.1\ \mu A$. $I_C \approx 4.71$ mA. $V_{CE} \approx 12-(4.71mA)(2k\Omega)=2.58$ V.

Stability

If β changes, I_C changes almost proportionally. Temperature-related leakage can also shift current. Fixed bias is easy to calculate but often unsuitable when strong stability is required.

> **Exam conclusion / points to remember:** Always check the calculated Q-point against the load line and operating region. Do not blindly use $I_C=\beta I_B$ if saturation would occur.

3. Collector-to-base feedback bias

Introduction — Collector-to-base feedback bias uses a resistor from collector to base to provide negative feedback.

Circuit action

When collector current increases, the voltage drop across R_C increases and collector voltage falls. Since the base resistor is connected to the collector, base current tends to decrease. This opposes the initial increase in collector current.

Why it stabilizes

This is negative feedback: the change in output is fed back in a way that reduces the original change. Compared with fixed bias, the operating point becomes less sensitive to β and temperature.

Analysis method

Write KVL around the base-emitter and collector loops for the exact schematic. Because the base resistor is connected to the collector, the voltage across it depends on collector voltage; this must be included. Use $I_E=I_C+I_B$ and the transistor active-region relation as appropriate.

Limitations

The feedback improves stability but does not make the circuit perfectly independent of transistor parameters. The operating point and gain depend on resistor values and the actual transistor.

Exam conclusion / points to remember: Explain the cause-and-effect sequence: I_C rises → collector voltage falls → I_B falls → I_C is pulled back.

4. Voltage-divider bias using Thevenin's theorem

Introduction — Voltage-divider bias uses two resistors to establish a base voltage and an emitter resistor to provide negative feedback. It is commonly used because it is more stable than fixed bias.

Divider voltage

For R_1 from V_{CC} to base and R_2 from base to ground, if divider current is sufficiently larger than base current, $V_B \approx V_{CC} R_2/(R_1+R_2)$.

Emitter current

For a silicon transistor approximation, $V_E \approx V_B - V_{BE}$ and $I_E \approx V_E/R_E$. Since I_C is close to I_E for large β, collector current is less dependent on β than in fixed bias.

Thevenin equivalent

Replace the divider by $V_{TH}=V_{CC} R_2/(R_1+R_2)$ and $R_{TH}=R_1 \| R_2$. The base loop gives $I_B \approx (V_{TH}-V_{BE})/(R_{TH}+(\beta+1)R_E)$.

Stabilization

If current rises, voltage across R_E rises and emitter voltage increases. That reduces V_{BE}, opposing further current increase. This is negative feedback.

Design check

Check that the divider current is large enough for the desired approximation, verify the Q-point, and ensure resistor power ratings and transistor limits are respected.

Exam conclusion / points to remember: Voltage-divider bias plus emitter resistance is a common stable bias arrangement. The Thevenin equation is useful when base current is not negligible.

5. Stability factor and thermal runaway

Introduction — Stability measures how much a transistor's operating current changes when temperature, leakage or transistor parameters change.

Causes of instability

β varies between devices and with current; leakage current increases with temperature; V_BE usually decreases as temperature rises. These effects can shift collector current and operating point.

Thermal runaway

Higher junction temperature can increase leakage and current. Higher current can increase transistor dissipation P≈V_CE I_C, raising temperature further. If the loop is not controlled, the device may exceed its safe operating area.

Stability factor

A stability factor is a sensitivity measure, such as S=∂I_C/∂I_CBO under specified fixed conditions. A lower magnitude generally indicates better stability for that mechanism. Different textbooks may define several stability factors for different varying parameters, so identify which is being used.

Prevention

Use emitter resistors, feedback bias, a well-designed divider, suitable heat sinks, thermal design, and operation within safe voltage/current/power limits. In power stages, thermal compensation may be required.

Exam conclusion / points to remember: Thermal runaway is positive feedback between temperature, current and power. Bias stabilization is essential for reliable operation.

UNIT IV — TRANSISTOR AMPLIFIERS

1. Small-signal amplification

Introduction — A transistor amplifier converts a small input signal into a larger output signal by controlling energy drawn from a DC supply.

Bias and signal

First establish a DC Q-point. The input signal then creates small changes around that point. Small-signal analysis treats these changes as incremental quantities, such as v_be and i_c, while DC values are represented separately.

Linear operation

For small enough changes, the nonlinear transistor characteristic can be approximated by a local straight line. This produces a linear model. If the signal is too large, the transistor may enter cutoff or saturation and the output becomes distorted.

Coupling and bypass capacitors

Coupling capacitors pass AC while blocking DC between stages. An emitter bypass capacitor reduces AC emitter degeneration over frequencies where its reactance is low, while the emitter resistor still helps set DC bias.

Gain definitions

EQ:A_v = v_o/v_i; A_i = i_o/i_i

Practical factors

Gain depends on transistor parameters, source resistance, load, frequency, coupling capacitors, bypass capacitors and output resistance. State whether a gain is voltage gain, current gain or power gain.

> **Exam conclusion / points to remember:** Explain that amplification transfers energy from the DC supply to the signal; the transistor does not create energy.

2. CE amplifier h-parameter equivalent circuit

Introduction — The h-parameter model simplifies a transistor's small-signal behavior into a two-port network around its bias point.

Two-port equations

EQ:v_i = h_ie i_i + h_re v_o

Output equation

EQ:i_o = h_fe i_i + h_oe v_o

Model elements

h_ie is represented by an input resistance; h_re couples output voltage back to input; h_fe represents controlled output current from input current; h_oe represents output conductance. The exact equivalent circuit depends on sign conventions and the selected transistor configuration.

Loaded stage

A load connected to the collector changes the effective AC load and therefore gain. Coupling capacitors are treated as AC shorts only in the midband approximation; supply rails are AC ground in a small-signal model.

Assumptions

The model is local to the operating point and frequency range. For a first approximation, h_re and h_oe may sometimes be neglected, but only when the problem permits.

Exam conclusion / points to remember: Draw the model with all four parameters identified, then write the two port equations before deriving gain.

3. CE voltage gain and phase inversion

Introduction — The common-emitter amplifier commonly provides substantial voltage gain and reverses the phase of the output voltage.

Cause of inversion

An increase in input/base current increases collector current. The larger current produces a larger voltage drop across the collector load, so collector voltage decreases. Therefore, output voltage changes in the opposite direction to input voltage.

Approximate gain

With emitter effectively AC-grounded and h_re/h_oe effects neglected, a common teaching approximation is $A_v \approx -h_{fe} R_C'/h_{ie}$, where R_C' is the effective AC collector load. The negative sign indicates phase inversion.

Intrinsic emitter resistance model

Another approximation for a bypassed emitter is $A_v \approx -R_C'/r_e$, where $r_e \approx V_T/I_E$ at room temperature under a simplified BJT model. The equations are not identical in all circuits; use the model specified by the question.

Load and source effects

A finite load reduces the effective collector resistance. Source resistance and input resistance determine how much source voltage reaches the transistor. Real gain is generally lower than a no-loading estimate.

Distortion

Incorrect biasing or excessive input amplitude can cause clipping. Frequency-dependent capacitors and transistor capacitances also make gain vary with frequency.

Exam conclusion / points to remember: Include the circuit, AC equivalent, gain derivation and a note that the minus sign indicates approximately 180° phase shift.

4. Effect of emitter bypass capacitor

Introduction — The emitter resistor stabilizes DC bias but also introduces AC negative feedback. A bypass capacitor can reduce this feedback at signal frequencies.

Without bypass

The emitter resistor carries signal current and develops an AC voltage. This reduces changes in base-emitter voltage and lowers gain, but improves linearity and can reduce sensitivity to transistor variation.

With bypass

At frequencies where capacitor reactance is small compared with the relevant resistance, the capacitor provides an AC path around the emitter resistor. The DC current still flows through the resistor, but AC gain increases.

Approximate formulas

EQ:$A_v \approx -R_C'/(r_e+R_E)$ (unbypassed simplified model)

Bypassed estimate

EQ:$A_v \approx -R_C'/r_e$ (bypass effective at signal frequency)

Frequency behavior

At low frequencies the capacitor reactance $X_C=1/(2\pi fC)$ is larger, so bypassing is less effective. At midband it may behave approximately as a short. At high frequencies, transistor internal capacitances and layout parasitics affect gain.

Exam conclusion / points to remember: The bypass capacitor trades some AC feedback for higher gain. It does not remove the need for a stable DC bias network.

5. CB and CC amplifier characteristics

Introduction — Common-base and common-collector stages offer different impedance and gain properties from the common-emitter stage.

Common-base

CB has low input resistance and high output resistance. Current gain is approximately α, near but below unity, while voltage gain may be high. It is useful at high frequencies and in some impedance-matching applications.

Common-collector

CC, or emitter follower, has high input resistance, low output resistance and voltage gain slightly below unity. It provides current gain and is useful as a buffer.

Comparison

CE is preferred for voltage gain and has phase inversion. CB is useful where low input resistance and high-frequency operation matter. CC is useful where load driving and impedance transformation matter.

Analysis

The h-parameter method can be used for each configuration, but parameters and terminal variables must be defined for that configuration. Do not copy CE equations unchanged into CB or CC analysis.

Exam conclusion / points to remember: Choose a configuration based on the required gain and input/output resistance rather than memorizing only one 'best' circuit.

UNIT V — SPECIAL DIODES, FETs AND ADVANCED DEVICES

1. SCR: construction, operation and characteristics

Introduction — An SCR is a four-layer PNPN semiconductor switching device with anode, cathode and gate terminals. It is used in controlled power conversion.

Structure

The PNPN structure forms three junctions. The outer terminals are anode and cathode; the gate is connected near the cathode-side region to trigger the device.

Forward blocking and triggering

With anode positive relative to cathode, the SCR can remain in forward blocking until triggered by a gate current or driven to breakover. Gate triggering initiates regenerative action and switches the device into conduction.

Latching and holding

After turn-on, anode current must rise above the latching current for the SCR to remain on after the gate signal is removed. Once conducting, current must fall below holding current to turn it off in a conventional SCR. The gate generally cannot turn off a standard SCR.

Characteristics and applications

The I–V curve includes reverse blocking, forward blocking and on-state conduction. SCRs are used in controlled rectifiers, motor control, heater control and power switching. They require attention to dv/dt, di/dt, gate ratings and commutation.

> **Exam conclusion / points to remember:** Differentiate latching current (needed immediately after turn-on) from holding current (minimum current to remain on).

2. Tunnel diode and negative differential resistance

Introduction — A tunnel diode is a heavily doped PN junction with a very thin depletion layer that allows quantum tunnelling.

Working principle

Because the depletion region is extremely thin, carriers can tunnel through the junction at low forward voltage. The current rises to a peak value as energy states align, then decreases as voltage increases further and alignment becomes less favorable.

Negative differential resistance

In the region between peak current and valley current, increasing voltage causes current to decrease. The differential resistance dV/dI is negative in that region. This is different from ordinary positive resistance behavior.

Characteristics and uses

The I–V curve is marked by peak voltage/current and valley voltage/current. Negative differential resistance can support oscillators, high-frequency circuits and specialized switching applications.

Limitations

Tunnel diodes require careful biasing and are not general-purpose replacements for ordinary rectifier diodes. Their behavior depends strongly on doping and device construction.

> **Exam conclusion / points to remember:** Clearly mark peak point, negative-resistance region and valley point on the I–V curve.

3. Varactor diode

Introduction — A varactor is a PN junction intentionally used as a voltage-controlled capacitor, normally under reverse bias.

Principle

The depletion region acts as the dielectric region between charge regions. Increasing reverse bias widens the depletion layer, so capacitance decreases.

Capacitance relation

A common model is C_j=C_0/(1+V_R/Φ)^m, where C_0 is zero-bias capacitance, V_R is reverse bias, Φ is a junction potential parameter and m depends on junction profile. This is a model, not a universal exact formula.

Applications

Varactors tune resonant circuits in radio receivers, voltage-controlled oscillators, frequency synthesizers and RF filters. A control voltage changes capacitance and therefore resonant frequency.

Design considerations

Observe reverse-voltage rating, capacitance range, Q factor, series resistance and parasitic capacitance. Avoid forward bias unless a specific circuit is designed for it.

> **Exam conclusion / points to remember:** More reverse bias generally means a wider depletion region and lower junction capacitance.

4. Photodiode and solar cell

Introduction — Photodiodes detect light, while solar cells convert light into useful electrical power. Both rely on photon-generated electron-hole pairs.

Photodiode operation

Incident photons create electron-hole pairs. The junction field separates carriers and produces photocurrent. Reverse bias is commonly used for fast detection because it widens the depletion region and reduces junction capacitance, although photovoltaic operation at zero bias is also possible.

Solar-cell operation

A solar cell is operated to deliver power to a load. Light-generated current shifts the diode I–V curve, producing a nonzero short-circuit current and open-circuit voltage. Maximum power is obtained at a point between these limits.

Key parameters

Photodiode response depends on wavelength, responsivity, dark current, bandwidth and noise. Solar-cell performance is described using open-circuit voltage, short-circuit current, fill factor and efficiency.

Applications

Photodiodes are used in optical communication, light meters, safety sensors and imaging. Solar cells are used in photovoltaic panels, calculators and remote power systems.

> **Exam conclusion / points to remember:** Photodiode: sensing signal from light. Solar cell: supplying power from light. Their device physics is related, but operating objectives differ.

5. LED and Schottky diode

Introduction — LEDs emit light under forward bias; Schottky diodes use a metal-semiconductor junction and are valued for fast switching and low forward voltage.

LED construction and working

An LED uses a semiconductor with a suitable bandgap. Forward current injects electrons and holes, which recombine and release energy as photons. The bandgap largely determines emitted wavelength and therefore color.

LED circuit requirements

LED current must be limited using a resistor or driver. A simple resistor estimate is R=(V_S–V_F)/I_F. Check the LED's maximum forward current, reverse-voltage rating, thermal conditions and optical requirements.

Schottky principle

A Schottky diode forms a rectifying metal-semiconductor barrier rather than a conventional PN junction. It has low charge storage and therefore fast switching, with a relatively low forward voltage in many applications.

Comparison and applications

LEDs are used for indicators, displays and lighting. Schottky diodes are used in fast rectifiers, switching power supplies and clamping circuits. Schottky devices can have higher reverse leakage, so temperature and leakage ratings matter.

> **Exam conclusion / points to remember:** LED: electrical-to-optical conversion. Schottky diode: low forward drop and fast switching, with leakage trade-offs.

6. JFET: structure, operation and characteristics

Introduction — A JFET is a voltage-controlled device in which a reverse-biased gate-channel junction controls the width of a conducting channel.

Structure

An N-channel JFET has an N-type channel with P-type gate regions. Source and drain connect to the channel; the gate-channel junction is reverse biased in normal operation. A P-channel JFET uses opposite polarities.

Operation

At small drain-source voltage, drain current rises approximately with V_DS. As V_DS increases, the depletion region widens near the drain and the channel narrows. At pinch-off, current approaches saturation for a fixed gate-source voltage. Making V_GS more negative in an N-channel JFET narrows the channel further and reduces I_D.

Shockley equation

EQ:I_D = I_DSS(1 – V_GS/V_P)^2

Parameter meaning

I_DSS is drain current at V_GS=0 in the specified saturation region. V_P is the pinch-off voltage with the usual negative sign for an N-channel JFET. At V_GS=V_P, the ideal equation predicts I_D=0.

Applications

JFETs are used in high-input-impedance stages, low-noise circuits and analog switching. The gate current is small, but the gate junction must not be forward biased beyond device limits.

> **Exam conclusion / points to remember:** Explain the output characteristic families and show how gate voltage controls drain current.

7. MOSFET: enhancement and depletion modes

Introduction — A MOSFET controls channel conductivity through an insulated gate. It is central to digital logic, switching power circuits and integrated circuits.

Structure

The gate is separated from the semiconductor by a thin insulating oxide. The terminals are gate, source, drain and often body. The insulated gate produces high input resistance, although gate capacitance and leakage remain important.

Enhancement NMOS

At V_GS below threshold V_TH, there is no strong inversion channel in the ideal model. When V_GS exceeds V_TH, an electron channel forms and drain current flows. In the long-channel model, the device has linear/triode and saturation regions.

Ideal saturation equation

EQ:I_D ≈ ½ μ_n C_ox(W/L)(V_GS–V_TH)^2

Assumptions

This expression applies to a long-channel enhancement NMOS in saturation with V_GS>V_TH and sufficient V_DS. It neglects channel-length modulation, mobility degradation, short-channel effects and other nonidealities.

Depletion MOSFET

A depletion-mode MOSFET has a channel at zero gate bias. Gate voltage can deplete or enhance the channel depending on polarity. This differs from an enhancement device, which requires the appropriate gate bias to induce the channel.

Applications and caution

MOSFETs are used in CMOS logic, power conversion and switching. Observe gate oxide voltage limits, ESD precautions, threshold variation and safe operating area.

> **Exam conclusion / points to remember:** Clearly distinguish a device's threshold condition and operating regions. State the long-channel assumptions when using the square-law equation.

8. FinFET structure and scaling

Introduction — A FinFET is a multi-gate transistor with a thin, fin-shaped channel. The gate controls the channel from multiple sides.

Structure

The channel rises vertically from the substrate like a fin. The gate wraps around the fin's two sides and top in a common tri-gate design, giving stronger electrostatic control than a single top gate.

Why it helps

As transistor dimensions shrink, a planar gate may struggle to control the channel, causing short-channel effects and leakage. A FinFET's multi-sided gate improves control over the channel potential and can reduce these effects.

Benefits and trade-offs

Potential benefits include improved switching control and scaling. Trade-offs include complex three-dimensional fabrication, fin geometry constraints, process variability, parasitic capacitances and increased manufacturing complexity.

Comparison

A FinFET is a transistor architecture, while CMOS describes complementary NMOS/PMOS circuit technology. FinFETs can be used to implement CMOS logic; the terms are not mutually exclusive categories.

> **Exam conclusion / points to remember:** Use a labelled cross-section showing fin, gate, source, drain and substrate.

9. CNTFET, ballistic transport and fabrication

Introduction — A carbon-nanotube field-effect transistor uses a semiconducting carbon nanotube as its conducting channel.

Structure

A CNTFET generally has source and drain contacts connected by a carbon nanotube channel and a gate that electrostatically controls channel conduction. The gate geometry can be top-gate, back-gate or surrounding gate depending on the design.

Transport

Ballistic transport means carriers traverse the channel with little or no scattering over the channel length. This can reduce some transport losses, but actual behavior depends on channel length, defects, contacts, temperature and scattering mechanisms.

Potential advantages

CNTFETs are researched for nanoscale devices, potentially strong electrostatic control and low-power switching. Their properties depend on nanotube diameter and chirality; only suitable semiconducting tubes are desired for logic.

Fabrication challenges

Challenges include controlled growth, placement and alignment, separating metallic from semiconducting nanotubes, contact resistance, reproducibility, contamination control and integration with existing semiconductor manufacturing.

Status and balanced conclusion

CNTFETs are promising research devices but are not a universal drop-in replacement for mainstream CMOS. A good answer should present both potential benefits and manufacturing barriers.

> **Exam conclusion / points to remember:** Define ballistic transport accurately and explain why chirality, contacts and integration are major challenges.

10. CMOS vs FinFET vs CNTFET

Introduction — CMOS, FinFET and CNTFET refer to related but different concepts in transistor and integrated-circuit technology.

CMOS

Complementary metal-oxide-semiconductor logic uses both NMOS and PMOS transistors. In a static CMOS inverter, one device is normally on and the other off in steady states, so ideal static power is low apart from leakage. CMOS is a circuit technology with a mature manufacturing and design ecosystem.

FinFET

FinFET is a transistor structure that improves gate control by surrounding a fin-shaped channel on multiple sides. FinFETs can be used in CMOS circuits; therefore CMOS and FinFET are not opposites.

CNTFET

CNTFET uses a semiconducting carbon nanotube channel. It is a candidate for future nanoscale devices, but manufacturing uniformity, metallic tubes, contacts and integration remain important challenges.

Comparison method

Compare channel material/geometry, gate control, manufacturing maturity, scalability, leakage behavior, potential performance and production challenges. Avoid unsupported claims that one technology is always faster or more efficient; results depend on process and circuit design.

> **Exam conclusion / points to remember:** Key distinction: CMOS is a complementary circuit technology; FinFET is a transistor architecture; CNTFET is a device using a carbon-nanotube channel.

SOLVED NUMERICALS AND DERIVATION PRACTICE

Numerical 1 — BJT current gain

Introduction — Use the relationship between α, β and terminal currents.

Problem

A transistor has $\alpha=0.97$ and emitter current $I_E=2$ mA. Find β, I_C and I_B.

Step 1

EQ:$\beta = \alpha/(1-\alpha) = 0.97/0.03 = 32.33$

Step 2

EQ:$I_C = \alpha I_E = 0.97 \times 2$ mA $= 1.94$ mA

Step 3

EQ:$I_B = I_E - I_C = 2.00 - 1.94 = 0.06$ mA $= 60\ \mu$A

Check

$I_C+I_B=1.94$mA$+0.06$mA$=2$mA, consistent with $I_E=I_C+I_B$.

Exam conclusion / points to remember: Keep current units consistent before using ratios.

Numerical 2 — Full-wave rectifier average output

Introduction — Use the ideal full-wave average output expression.

Problem

An ideal full-wave rectifier has peak voltage $V_m=18$ V. Find the average output.

Formula and substitution

EQ:$V_{DC} = 2V_m/\pi = 2\times18/\pi \approx 11.46$ V

Interpretation

This is the unfiltered ideal average. A capacitor filter changes the average and ripple; practical diode drops and source resistance also affect the output.

Exam conclusion / points to remember: State whether the supplied peak is across the full secondary or a half-secondary.

Numerical 3 — JFET drain current

Introduction — Apply the JFET Shockley equation within its valid region.

Problem

$I_{DSS}=8$ mA, $V_P=-4$ V and $V_{GS}=-2$ V. Estimate I_D.

Calculation

EQ:$I_D = I_{DSS}(1-V_{GS}/V_P)^2 = 8(1-0.5)^2 = 2$ mA

Check

Since V_{GS} is halfway between 0 V and −4 V, the ideal square-law model gives one-quarter of I_{DSS}.

Exam conclusion / points to remember: Check signs carefully: $(-2)/(-4)=+0.5$.

Numerical 4 — Capacitor-filter ripple

Introduction — Estimate the peak-to-peak ripple for a full-wave rectifier with a capacitor filter.

Problem

A full-wave rectifier operates from 50 Hz AC, draws 100 mA load current and uses a 1000 μF capacitor. Estimate ripple.

Step 1

EQ:f_r = 2f = 100 Hz; C = 1000 μF = 0.001 F

Step 2

EQ:V_r(pp) ≈ I_L/(f_r C) = 0.1/(100×0.001) = 1 V

Interpretation

The result is approximate and assumes relatively small ripple. Practical ripple depends on transformer impedance, capacitor ESR and conduction angle.

Exam conclusion / points to remember: Convert microfarads to farads before substituting.

Numerical 5 — LED series resistor

Introduction — Choose a current-limiting resistor for a simple LED circuit.

Problem

A 5 V supply drives an LED with forward voltage 2 V at 10 mA. Estimate the series resistor.

Calculation

EQ:R = (V_S–V_F)/I_F = (5–2)/0.010 = 300 Ω

Practical selection

A standard resistor near 300 Ω may be selected after checking actual LED forward voltage, supply tolerance and desired current. Resistor power is P=I²R=0.01²×300=0.03 W, so choose an appropriately rated resistor with margin.

Exam conclusion / points to remember: Current limiting protects the LED. Do not connect an ordinary LED directly to a voltage source without a suitable driver.

UNIT-WISE LONG-ANSWER QUESTION BANK

Unit I — Diode characteristics and applications

1. Explain intrinsic and extrinsic semiconductors and the effect of doping.
2. Explain PN junction formation, depletion region and forward/reverse bias.
3. Draw and explain diode I–V characteristics; compare ideal and practical models.
4. Explain diode static/dynamic resistance and junction/diffusion capacitance.
5. Explain half-wave rectifier operation, waveforms and average output derivation.
6. Compare centre-tapped and bridge full-wave rectifiers.
7. Explain capacitor filter operation and derive the approximate ripple expression.
8. Explain clipper and clamper circuits and compare their functions.
9. Explain Zener diode voltage regulation and discuss worst-case design checks.

Unit II — BJT

1. Explain BJT construction and transistor action with a labelled diagram.
2. Derive the relationship between α and β and solve a current-gain numerical.
3. Explain CB configuration and its input/output characteristics.
4. Explain CE configuration, characteristics and operating regions.
5. Explain CC configuration and why it is called an emitter follower.
6. Compare CB, CE and CC configurations.
7. Explain the CE h-parameter model and determine parameters from characteristics.

Unit III — Biasing

1. Explain the need for transistor biasing and define Q-point.
2. Draw the DC load line and explain its intercepts and operating point.
3. Derive fixed-bias equations and calculate the Q-point.
4. Explain collector-to-base feedback bias and its stabilizing action.
5. Explain voltage-divider bias using Thevenin's theorem.
6. Define stability factor and explain thermal runaway and its prevention.

Unit IV — Amplifiers

1. Explain the transistor as a small-signal amplifier.
2. Draw and explain the CE h-parameter equivalent circuit.
3. Derive approximate CE voltage gain and explain phase inversion.
4. Explain the role of coupling and emitter bypass capacitors.
5. Compare CE, CB and CC amplifier characteristics.
6. Explain why practical amplifier gain varies with load and frequency.

Unit V — Special devices and FETs

1. Explain SCR structure, forward blocking, triggering, latching and holding current.
2. Explain tunnel diode characteristics and negative differential resistance.
3. Explain varactor diode operation as a voltage-controlled capacitor.
4. Compare photodiode and solar cell operation and applications.
5. Explain LED construction, working and current limiting.
6. Explain Schottky diode operation, advantages and limitations.
7. Explain JFET structure, operation, characteristics and Shockley equation.
8. Compare enhancement and depletion MOSFETs.
9. Explain FinFET structure and its benefits for scaling.
10. Explain CNTFET structure, ballistic transport and fabrication challenges.
11. Compare CMOS, FinFET and CNTFET technologies.

FINAL REVISION: FORMULAS AND EXAM CHECKLIST

Diode / rectifier

$I_D = I_S[\exp(V_D/(nV_T)) - 1]$

$r_d \approx nV_T/I_D$ (forward bias, ideal exponential model)

Half-wave: $V_{DC}=V_m/\pi$; Full-wave: $V_{DC}=2V_m/\pi$

Ripple: $V_{r(pp)}\approx I_L/(f_r C)$; $f_r=f$ (half-wave), $2f$ (full-wave)

Zener: $R_S=(V_S-V_Z)/(I_L+I_Z)$; $P_Z=V_Z I_Z$

BJT / bias

$I_E=I_C+I_B$; $\alpha=I_C/I_E$; $\beta=I_C/I_B$

$\beta=\alpha/(1-\alpha)$; $\alpha=\beta/(\beta+1)$

Fixed bias: $I_B\approx(V_{CC}-V_{BE})/R_B$; $V_{CE}\approx V_{CC}-I_C R_C$

Load line: $I_C=(V_{CC}-V_{CE})/R_C$

Divider: $V_{TH}=V_{CC}\, R_2/(R_1+R_2)$; $R_{TH}=R_1||R_2$

$I_B\approx(V_{TH}-V_{BE})/(R_{TH}+(\beta+1)R_E)$

Amplifier / FET

CE h-model: $v_i=h_{ie} i_i+h_{re} v_o$; $i_o=h_{fe} i_i+h_{oe} v_o$

CE gain estimate: $A_v\approx -R_C'/r_e$ (emitter bypass effective)

JFET: $I_D=I_{DSS}(1-V_{GS}/V_P)^2$ (valid region only)

Long-channel NMOS saturation: $I_D\approx \tfrac{1}{2}\mu_n C_{ox}(W/L)(V_{GS}-V_{TH})^2$

Before submitting an exam answer

- Have I defined the main term accurately?
- Have I drawn a labelled diagram or characteristic curve if the question needs one?
- Have I explained construction/principle and working in a logical order?
- Have I defined symbols and stated model assumptions?
- Have I shown the important intermediate derivation steps?
- Are numerical answers in correct units and reasonable magnitude?
- Have I included applications, limitations or comparison points where relevant?
- Have I avoided treating model questions as guaranteed or previous-year questions?

Technical references

Primary scope reference: JNTUH B.Tech CSE R25 Course Structure & Syllabus, EC104ES Electronic Devices and Circuits. Recommended supporting texts include Boylestad & Nashelsky, Electronic Devices and Circuit Theory; Sedra & Smith, Microelectronic Circuits; Millman & Halkias, Electronic Devices and Circuits; and Neamen, Electronic Circuit Analysis and Design. Follow the exact textbook edition and sign convention prescribed by your instructor.

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